Alex Pedrini

PhD student in Astronomy at Stockholm University
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EXPERIENCE

Stockholm University, Stockholm — *PhD student*

AUGUST 2022 - PRESENT

PhD candidate in astronomy focused on star formation and stellar feedback from emerging young star clusters. Active member of the FEAST program - a JWST survey investigating stellar feedback processes in six nearby galaxies.

Thesis supervisors: Angela Adamo (Stockholm University), Arjan Bik (Stockholm University), Daniela Calzetti (UMass Amherst).

EDUCATION

Stockholm University, Stockholm — *Licenciate*

AUGUST 2022 - SEPTEMBER 2024

Licentiate's degree in Astronomy, September 26th, 2024; mark: *Pass*

Title: *The emergence phase of star formation as a key probe for feedback timescales*

Supervisors: Angela Adamo (Stockholm University), Arjan Bik (Stockholm University), Daniela Calzetti (UMass Amherst).

Università degli studi di Milano Bicocca, Milano, Italy — *Master's degree*

OCTOBER 2019 - OCTOBER 2021

Master's degree in Astrophysics and Space Physics, October 26th, 2021; mark: 110/110.

Title: *Towards a complete view of ram pressure stripping in the galaxy cluster A1367*

Supervisors: Michele Fumagalli (UniMiB), Matteo Fossati (UniMiB)

Università degli studi di Milano Bicocca, Milano, Italy — *Bachelor's degree*

OCTOBER 2016 - SEPTEMBER 2019

Bachelor's degree in Physics, September 19th, 2019; mark: 101/110.

SKILLS

Programming & Scripting: Python, C/C++, Bash

Software & Tools: Astrodendro, SAOImageDS9, CIGALE, SLUG, Jupyter Notebook, TOPCAT, APT and ETC (JWST), PaKo

Web development: Jekyll, GitHub Pages, HTML, Markdown

Operating systems: macOS, Linux, Windows

AWARDS

Stiftelsen Hierta Retzius fond för vetenskaplig forskning, Kungl. Vetenskapsakademien, travel grant, 65000 SEK

2025 Spring Symposium, STScI, travel grant, 1500 USD

Dahlmark 2025, Department of Astronomy, Stockholm University, travel grant, 10000 SEK

Dahlmark 2024 II, Department of Astronomy, Stockholm University, travel grant, 7000 SEK

Dahlmark 2024 I, Department of Astronomy, Stockholm University, travel grant, 5000 SEK

Dahlmark 2023, Department

Title: *Misure della conducibilità termica di un campione di assorbitore per microonde a temperature criogeniche (Thermal conductivity measurements of a sample of microwave absorber at cryogenic temperatures)*

Supervisors: Mario Zannoni (UniMiB)

PUBLICATIONS

Corresponding author

1. **Pedrini A.**, Adamo A., Calzetti D., et al., submitted

   Mass regulates the emerging timescales of young star clusters

2. **Pedrini A.**, Adamo A., Bik A., et al., 2024, ApJ, 992, 96P

   The near infrared SED of emerging young star clusters in the FEAST galaxies: Missing ingredients at 1 – 5 μm

3. **Pedrini A.**, Adamo A., Calzetti D., et al., 2024, ApJ, 973, 67P

   Erratum: "Feedback in Emerging extragAlactic Star clusTers: JWST spots polycyclic aromatic hydrocarbon destruction in NGC 628 during the emerging phase of star formation" (2024, ApJ, 971, 32)

4. **Pedrini A.**, Adamo A., Calzetti D., et al., 2024, ApJ, 971, 32P

   Feedback in Emerging extragAlactic Star clusTers: JWST spots polycyclic aromatic hydrocarbon destruction in NGC 628 during the emerging phase of star formation

5. **Pedrini A.**, Fossati M., Gavazzi G., et al., 2022, MNRAS, 511, 5180P

   MUSE sneaks a peek at extreme ram-pressure stripping events - V. Towards a complete view of the galaxy cluster A1367

Others:

1. Bortolini G., Correnti M, ..., **Pedrini A.,** et al., 991, 212B

   FEAST: JWST/NIRCam view of the resolved stellar populations of the interacting dwarf galaxies NGC 4485 and NGC 4490

2. Correnti M., Bortolini G., ..., **Pedrini A.,** et al., 990, 72C

   FEAST: probing the stellar population of the starburst dwarf galaxy NGC4449 with JWST/NIRCam

3. Knutas A., Adamo A., **Pedrini A.,** et al., ApJ, 993, 13K

   FEAST: JWST uncovers the emerging timescales of young star clusters in M83

4. Draine B. T., ..., **Pedrini A.,** et al., ApJ, 984, 42D

   Detection of deuterated hydrocarbon nanoparticles in the Whirlpool galaxy, M51

5. Elmegreen B., ..., **Pedrini A.,** et al., OJAp, 8E, 21E

of Astronomy, Stockholm University, travel grant, 10500 SEK

Best Poster Award, Astronomdagarna 2024, Lund
Feedback in Emerging extragAlactic Star clusTers: first results

LANGUAGES

Italian (mother language), English (fluent), Swedish (good)

Power spectra of JWST images in Local Galaxies: searching for disk thickness

6. Calzetti D., ..., **Pedrini A.,** et al., ApJ, 971, 118C

   JWST-FEAST: Feedback in Emerging extrAgalactic Star clusTers: Calibration of star formation rates in the mid-infrared with NGC 628

7. Gregg B., ..., **Pedrini A.,** et al., ApJ, 971, 115G

   Feedback in Emerging extrAgalactic Star clusTers, FEAST: The relation between 3.3 μm Polycyclic Aromatic Hydrocarbon Emission and star formation rate traced by ionized gas in NGC 628

TALKS & POSTERS

Talks

The emerging sequence of young star clusters in local galaxies - UMASS invited seminar, Amhrest, USA - Nov 12, 2025

The morphological evolution of dusty star-forming regions in NGC 628 - EAS Annual Meeting, Cork, Ireland - Jun 27, 2025

A census of emerging young star clusters in nearby galaxies - EAS Annual Meeting, Cork, Ireland - Jun 24, 2025

Dissecting the SED of emerging young star clusters - STScI Spring Symposium: Inter+Stellar, Baltimore, MD, USA - May 15, 2025

The emergence phase of star formation as a key probe for feedback timescales - SU-UU meeting, Uppsala, Sweden - Nov 1, 2024

(Outreach talk) Stars in the making: a journey through galaxies with the JWST- Astronomins dag och natt 2024, Stockholm, Sweden - Oct 19, 2024

PAH destruction in NGC628 during the emergence phase of star formation - The physics and impact of astrophysical dust, Aspen, CO, USA - Mar 4, 2024

Posters

Feedback in Emerging extragAlactic Star clusTers: first results - Astronomdagarna 2024, Lund, Sweden - Oct 3, 2024

JWST spots PAH destruction from emerging star clusters in NGC 628 - EAS Annual Meeting, Padova, Italy - Jul 1, 2024

The morphology of dusty star-forming regions as revealed by JWST in NGC 628 - Olympian Symposium, Paralia Katerini, Greece - May 29, 2023